Groundswell: Components Map

Live app: https://devpulse-sable-seven.vercel.app · **Diagram:** https://devpulse-sable-seven.vercel.app/architecture
Repository: https://github.com/niralidesai09/devpulse (branch `roadmap`)
Download: https://devpulse-sable-seven.vercel.app/groundswell-components-map.pdf

Groundswell is a weekly developer-community intelligence digest for Sumit, the PM of a code review automation startup. Every Friday at 08:00 IST it collects live discussions, analyses them with **Google Gemini Flash**, and emails an evidence-linked brief.

Data flow (one weekly run)

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Vercel Cron (Fri 02:30 UTC = 08:00 IST)  or  "Run now" (admin passcode)
        │
        ▼
/api/cron/weekly  →  runPipeline()   (one Vercel Function, ~2 min, limit 300 s)
        │
        ├─ 1. Load active topics + keywords ........................... Supabase
        ├─ 2. Collect last-7-day posts (parallel, dedupe, top 300) ..... HN Algolia · Stack Exchange · GitHub · DEV.to
        ├─ 3a. Classify posts in batches of 25 ......................... Gemini Flash
        ├─ 3b. (in parallel) keyword counts this week vs last week ..... HN Algolia  (smart baseline)
        ├─ 4. Save posts; compute trend % + Opportunity Score (code) ..... Supabase
        ├─ 5. Emerging Topic Radar: cluster "other" posts ................ Gemini Flash → suggested_topics
        ├─ 6. Write the 8-section Friday brief ......................... Gemini Flash → briefs
        ├─ 7. Email the brief ..................................... Resend → BRIEF_TO_EMAIL
        └─ 8. Mark run success / partial / failed (+ per-source status) .. Supabase

Dashboard (Next.js on Vercel) reads Supabase  ·  Ask the Pulse: question → /api/ask → search_posts() (Postgres full-text) → Gemini Flash → answer with links
```

Components

Triggers

Component	File	What it does	Service
Weekly cron	`vercel.json` → `app/api/cron/weekly/route.ts`	Starts the pipeline every Friday 02:30 UTC (08:00 IST). Requires `Authorization: Bearer CRON_SECRET`.	Vercel Cron
Run now	`components/run-now.tsx` → `app/api/run/route.ts`	Starts the same pipeline on demand (live demo).	Vercel Functions
Admin passcode	`lib/admin.ts`, `app/actions.ts`	Unlocks Run now, topic editing and radar approvals (httpOnly HMAC cookie, re-checked on the server).	Next.js Server Actions

Live data collection (`lib/sources`)

Component	File	What it does	Service
Hacker News adapter	`hackernews.ts`	Stories + comments per keyword, last 7 days.	HN Algolia API
Stack Overflow adapter	`stackoverflow.ts`	All questions of the last 7 days, matched to keywords locally (saves quota).	Stack Exchange API v2.3
GitHub Issues adapter	`github.ts`	Issues matching topic keywords (≤ 6 per OR query), bot authors skipped; open/closed kept as evidence.	GitHub Search API
DEV.to adapter	`devto.ts`	Newest 1,000 articles from 10 relevant tags, matched to keywords.	DEV.to (Forem) API
Collector	`index.ts`	Runs adapters in parallel, dedupes by normalised URL, keeps the 300 most-engaged posts, records per-source status. A failing source never stops the run.	—

AI: Google Gemini Flash only

Component	File	What it does	Service
Gemini client	`lib/ai/gemini.ts`	Zod schema → JSON Schema (`responseJsonSchema`), reply validated with Zod, 1 retry on invalid JSON, backoff on 429/5xx. Model from `GEMINI_MODEL`.	Gemini Flash via `@google/genai`

Component	File	What it does	Service
Classifier	`lib/ai/classify.ts`	Per post: topic / "other" / irrelevant, one-line pain point, unresolved + reason (uses source signals), sentiment −1..1, severity 1–5, competitors.	Gemini Flash
Emerging Topic Radar	`lib/pipeline/radar.ts`	Clusters "other" posts into ≤ 4 suggested topics; code requires ≥ 2 real threads, no duplicate names, ≥ 3 new keywords.	Gemini Flash
Brief writer	`lib/pipeline/brief.ts`	8-section brief. Tables are rendered by code; Gemini cites `[E#]` evidence ids that code converts into real links; numbers self-checked.	Gemini Flash
Ask the Pulse	`lib/ai/ask.ts`, `app/api/ask/route.ts`	Answers only from posts found by full-text search, with `[S#]` links; "not enough data" otherwise.	Gemini Flash + Supabase

Analysis (code, no AI)

Component	File	What it does	Service
Smart baseline	`lib/pipeline/baseline.ts`	Exact-phrase keyword counts for this week and the previous 7 days, so trends work from the very first run.	HN Algolia API
Scoring	`lib/pipeline/scoring.ts`	Hacker News keyword trend with counts ("too few to compare" under 5 matches), comparison with the previous comparable weekly run, and the relative Opportunity Score (top topic = 100).	—
Pipeline	`lib/pipeline/run.ts`	Orchestrates one run and records timings, errors and status.	Vercel Functions

Storage, email, UI

Component	File	What it does	Service
Database	`supabase/migrations/*.sql`, `lib/db/*`	`topics`, `runs`, `posts` (+ full-text index), `topic_stats`, `briefs`, `suggested_topics`. RLS on: public read-only; writes only on the server with the service-role key.	Supabase Postgres
Email	`lib/email/brief-email.ts`	Sends the brief (HTML + text) from `onboarding@resend.dev` to `BRIEF_TO_EMAIL`; stores `emailed_at` or the error.	Resend
App shell	`app/layout.tsx`, `components/shell/*`	Sidebar navigation (drawer on mobile), data freshness and the next scheduled run.	Next.js on Vercel

Component	File	What it does	Service
Overview `/`	`app/page.tsx`	Key figures; the brief's three answers (most discussed with counts, trending up or down vs last week, unresolved / underserved); the lead problem with evidence; top opportunities; week-over-week chart; the Friday brief.	Next.js on Vercel, Recharts
Trends `/trends`	`app/trends/page.tsx`, `components/charts.tsx`	Weekly volume, mentions by topic, Hacker News keyword trend, resolved vs unresolved, source contribution; every chart has a table view.	Next.js on Vercel, Recharts
Opportunities `/opportunities`	`app/opportunities/page.tsx`	Ranked topics: observed facts, labelled AI note, "Why this score?", evidence; Emerging Topic Radar; competitor and objection radar.	Next.js on Vercel
Evidence `/evidence`	`app/evidence/page.tsx`, `lib/db/evidence.ts`	Searchable, filterable feed of collected discussions with source, published and retrieved times.	Next.js on Vercel, Supabase
Ask `/ask`	`app/ask/page.tsx`, `components/ask-pulse.tsx`	Ask the Pulse chat with citations.	Next.js on Vercel, Gemini Flash
`/topics`	`app/topics/page.tsx`	Add / edit / pause topics and keywords (passcode).	Next.js on Vercel
`/runs`	`app/runs/page.tsx`	Run history with per-source status, brief and email status.	Next.js on Vercel
How it works `/architecture`	`app/architecture/page.tsx`	This map, the diagram, and download links (PDF / Markdown).	Next.js on Vercel

Quality checks

Component	File	What it does	Service
Unit tests	`tests/*.test.ts` (`npm test`)	31 tests: normalisation, dedupe, attribution, evidence links, metrics, zero baselines, AI schema validation, invalid URLs, partial runs.	Vitest
Provenance check	`scripts/verify-provenance.mjs` (`npm run verify:provenance`)	Re-fetches sampled posts from the official source APIs and compares id, title, time and URL.	HN, GitHub, Stack Exchange, DEV.to APIs
Data audit	`docs/DATA_AUDIT.md`	Source verification table, metric definitions, findings and limits.	—

External services

Service	Used for	Key / config
Google Gemini Flash	All AI tasks	`GEMINI_API_KEY`, `GEMINI_MODEL=gemini-3.8-flash`
Supabase	Postgres database, full-text search	`NEXT_PUBLIC_SUPABASE_URL`, `NEXT_PUBLIC_SUPABASE_ANON_KEY`, `SUPABASE_SERVICE_ROLE_KEY`
Hacker News (Algolia)	Posts + smart baseline	none
Stack Exchange API	Posts	`STACKEXCHANGE_KEY` (optional)
GitHub Search API	Issues	`GITHUB_TOKEN`
DEV.to API	Articles	none
Resend	Email delivery	`RESEND_API_KEY`, `BRIEF_TO_EMAIL`
Vercel	Hosting + Cron	`CRON_SECRET`, `ADMIN_PASSCODE`

All AI tasks use Google Gemini Flash via @google/genai. Live data only. Topics and keywords are stored in Supabase. Deployed on Vercel with a Friday 08:00 IST cron.